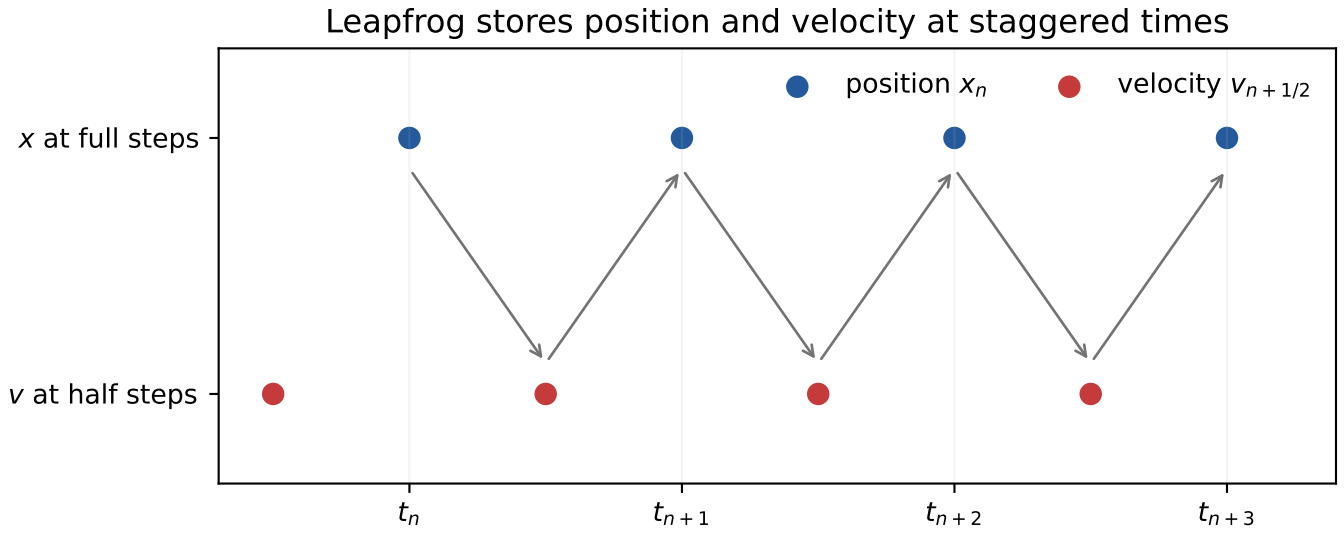
Leapfrog stores position and velocity at staggered times
position x_n
velocity $v_{n+1/2}$
x at full steps
v at half steps
t_n
t_{n+1}
t_{n+2}
t_{n+3}